

Quantum Computing and Option Pricing

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Grenoble INP - Ensimag, UGA

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Outline

1 Encoding a function in a quantum state

Current checkpoint

- Principle: construct an operator F on $n + 1$ qubits such that for all $i \in \{0, \dots, 2^n - 1\}$,
$$F \cdot (|i\rangle \otimes |0\rangle) = |i\rangle \otimes \left(\sqrt{1 - f_c(i)}|0\rangle + \sqrt{f_c(i)}|1\rangle \right)$$

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$$\begin{aligned} F \cdot (D_n \otimes \mathbf{I}) \cdot |0^{n+1}\rangle &= F \cdot [(D_n \cdot |0^n\rangle) \otimes |0\rangle] \\ &= F \cdot \left[\left(\sum_{w \in \mathbb{Z}_2^n} \sqrt{p_w} |w\rangle \right) \otimes |0\rangle \right] \end{aligned}$$

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QAE can be used to compute the probability of measuring “1” on the last qubit, this probability is $\sum_{i=0}^{2^n-1} p_i f_c(i) = \mathbb{E}[f_c(X)]$

A problem

Question

What generic technique can be used to store elements of the form $\sqrt{f(i)}$ in the amplitudes of a state?

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- **Approximation:** instead of storing elements of the form $\sqrt{1 - f_c(i)}|0\rangle + \sqrt{f_c(i)}|1\rangle$, we will store approximations of the form $\cos(\theta_i)|0\rangle + \sin(\theta_i)|1\rangle$

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- **Cost:** a loss in precision, the global algorithm will have an error of order $O\left(M^{-\frac{2}{3}}\right)$ instead of $O\left(M^{-1}\right)$

Approximating a function with rotations

Using the fact that $\sin^2\left(y + \frac{\pi}{4}\right) = y + \frac{1}{2} + O(y^3)$:

- $y \leftarrow d \cdot (f_c(X) - \frac{1}{2})$: we estimate $\mathbb{E}\left[\sin^2\left(d \cdot (f_c(X) - \frac{1}{2}) + \frac{\pi}{4}\right)\right]$ instead of $d \cdot \mathbb{E}[f_c(X)] + \frac{1-d}{2}$, where $d > 0$ is used to control the approximation error

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New task

Construct an operator F' such that for all $i \in \{0, \dots, 2^n - 1\}$,

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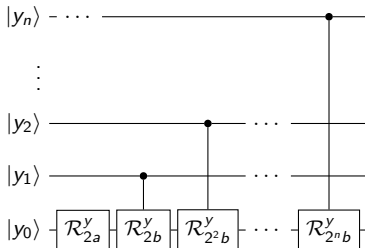
Nb: $f_c : i \mapsto \frac{(s_i - s_L)_+}{s_M - s_L}$ is piecewise linear, we focus on the encoding of linear functions

Approximating a linear function

We consider a function $g : \begin{cases} \{0, \dots, 2^n - 1\} & \rightarrow [0, 1] \\ x & \mapsto a + bx \end{cases}$ to approximate using rotations

We define the following operators on $n + 1$ qubits:

$$T_0^g \stackrel{\text{def}}{=} \mathcal{R}_{2a}^y, \quad T_k^g \stackrel{\text{def}}{=} C_k \left[\mathcal{R}_{2^k b}^y[0] \right] \text{ for } k = 1, \dots, n \text{ and } C^g \stackrel{\text{def}}{=} \prod_{k=n}^0 T_k^g$$

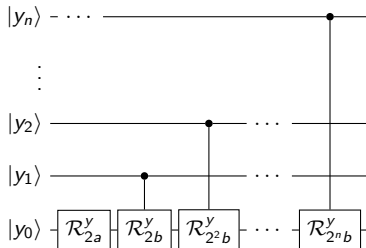


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Proposition

For all $x \in \{0, \dots, 2^n - 1\}$, we have $C^g \cdot (|x\rangle \otimes |0\rangle) = |x\rangle \otimes [\cos(g(x))|0\rangle + \sin(g(x))|1\rangle]$

Proof of the approximation

The proof is based on the following property (see the exercise sheet)

Proposition

If $|x\rangle = |x_{n-1} \cdots x_0\rangle$ and $0 \leq m \leq n-1$ then for all angles α and β ,

$$C_{m+1} [\mathcal{R}_\alpha^\gamma[0]] \cdot \left[|x\rangle \otimes \left(\cos\left(\frac{\beta}{2}\right) |0\rangle + \sin\left(\frac{\beta}{2}\right) |1\rangle \right) \right] = \\ |x\rangle \otimes \left(\cos\left(\frac{\alpha x_m + \beta}{2}\right) |0\rangle + \sin\left(\frac{\alpha x_m + \beta}{2}\right) |1\rangle \right)$$

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In particular, when $|x\rangle = |x_{n-1} \cdots x_0\rangle$ we have

$$T_{j+1}^g \cdot (|x\rangle \otimes (\cos(\alpha)|0\rangle + \sin(\alpha)|1\rangle)) = C_{j+1} [\mathcal{R}_{2^{j+1}b}^y[0]] \cdot (|x\rangle \otimes (\cos(\alpha)|0\rangle + \sin(\alpha)|1\rangle)) \\ = |x\rangle \otimes (\cos(\alpha + 2^j x_j b) |0\rangle + \sin(\alpha + 2^j x_j b) |1\rangle)$$

Proof of the approximation (cont'd)

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- **Note:** the proof can be adapted straightforwardly to show that, for all angles θ ,

$$C^g \cdot (|x\rangle \otimes (\cos(\theta)|0\rangle + \sin(\theta)|1\rangle)) = |x\rangle \otimes [\cos(g(x) + \theta) |0\rangle + \sin(g(x) + \theta) |1\rangle]$$

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Corollary

If g and h are linear functions then for all $x \in \{0, \dots, 2^n - 1\}$, we have

$$C^h \cdot C^g \cdot (|x\rangle \otimes |0\rangle) = |x\rangle \otimes [\cos(g(x) + h(x))|0\rangle + \sin(g(x) + h(x))|1\rangle]$$

Encoding the modified payoff

- **Goal:** encode $g_c : i \mapsto d \left(f_c(i) - \frac{1}{2} \right) + \frac{\pi}{4}$ using rotations
- **Assumption:** $K = s_L$ for some $L \in \{0, \dots, 2^n - 1\}$
- **Principle:**
 - ▶ If $i < L$ then $g_c(i) = \frac{\pi}{4} - \frac{d}{2}$
 - ▶ Otherwise we have

$$\begin{aligned} g_c(i) &= d \left(\frac{s_i - s_L}{s_M - s_L} - \frac{1}{2} \right) + \frac{\pi}{4} \\ &= \frac{d(s_M - s_m)}{(s_M - s_L)(2^n - 1)}(i - L) + \frac{\pi}{4} - \frac{d}{2} \end{aligned}$$

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Summary

Let $f_0 : i \mapsto \frac{\pi}{4} - \frac{d}{2}$, $f_1 : i \mapsto \frac{d(s_M - s_m)}{(s_M - s_L)(2^n - 1)}(i - L)$ and $r \stackrel{\text{def}}{=} \begin{cases} 0 & \text{if } i < L \\ 1 & \text{otherwise} \end{cases}$

Then $g_c(i) = f_0(i) + r \cdot f_1(i)$

A comparison operator

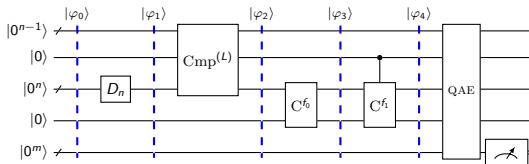
- The definition of g_c requires determining whether $i < L$ or not, we need to be able to construct a circuit that permits to compare integers
- **Principle:** for integers $n \geq 1$ and $L \in \{0, \dots, 2^n - 1\}$, consider an operator $\text{Cmp}^{(L)}$ such that

$$\text{Cmp}^{(L)} \cdot (|0^n\rangle \otimes |x\rangle) = \begin{cases} |s_x\rangle \otimes |0\rangle \otimes |x\rangle & \text{if } x < L \\ |s_x\rangle \otimes |1\rangle \otimes |x\rangle & \text{if } x \geq L \end{cases}$$

where each s_x is of length $n - 1$

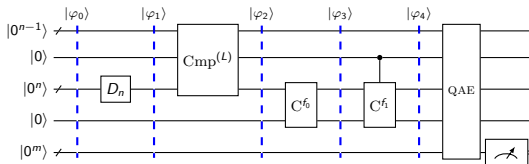
- A construction of (a standard version of) $\text{Cmp}^{(L)}$ is given in the exercise sheet

Wrapping it all up



$$|\varphi_0\rangle = |0^{2n+m+1}\rangle$$

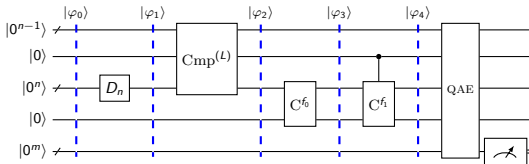
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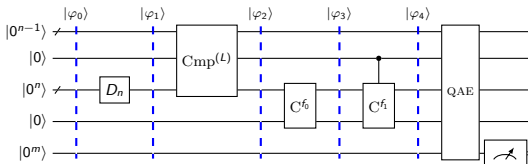


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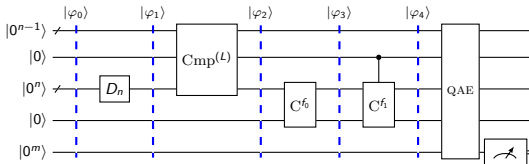
$$|\varphi_2\rangle = \left[\sum_{i=0}^{L-1} \sqrt{p_i} \cdot (|s_i\rangle \otimes |0\rangle \otimes |i\rangle \otimes |0\rangle) + \sum_{i=L}^{2^n-1} \sqrt{p_i} \cdot (|s_i\rangle \otimes |1\rangle \otimes |i\rangle \otimes |0\rangle) \right] \otimes |0^m\rangle$$

Wrapping it all up



$$|\varphi_3\rangle = \left[\sum_{i=0}^{L-1} \sqrt{p_i} \cdot (|s_i\rangle \otimes |0\rangle \otimes |i\rangle \otimes (\cos(f_0)|0\rangle + \sin(f_0)|1\rangle)) + \sum_{i=L}^{2^n-1} \sqrt{p_i} \cdot (|s_i\rangle \otimes |1\rangle \otimes |i\rangle \otimes (\cos(f_0)|0\rangle + \sin(f_0)|1\rangle)) \right] \otimes |0^m\rangle$$

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$$|\varphi_4\rangle = \left[\sum_{i=0}^{L-1} \sqrt{p_i} \cdot (|s_i\rangle \otimes |0\rangle \otimes |i\rangle \otimes (\cos(f_0)|0\rangle + \sin(f_0)|1\rangle)) + \sum_{i=L}^{2^n-1} \sqrt{p_i} \cdot (|s_i\rangle \otimes |1\rangle \otimes |i\rangle \otimes (\cos(f_0 + f_1(i))|0\rangle + \sin(f_0 + f_1(i))|1\rangle)) \right] \otimes |0^m\rangle$$

The probability of measuring 1 on the last qubit of φ_4 with a precision of m bits is

$$P_1 = \sum_{i=0}^{L-1} (p_i \cdot \sin^2(f_0)) + \sum_{i=L}^{2^n-1} (p_i \cdot \sin^2(f_0 + f_1(i)))$$

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 &= \sum_{i=0}^{L-1} (p_i \cdot \sin^2(\frac{\pi}{4} - \frac{d}{2})) + \sum_{i=L}^{2^n-1} (p_i \cdot \sin^2(d \cdot (\frac{i-L}{2^n-1-L} - \frac{1}{2}) + \frac{\pi}{4}))
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 &= \sum_{i=0}^{2^n-1} \left(p_i \cdot \sin^2 \left(d \cdot \left(\frac{(i-L)_+}{2^n-1-L} - \frac{1}{2} \right) + \frac{\pi}{4} \right) \right)
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The probability of measuring 1 on the last qubit of φ_4 with a precision of m bits is

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 \end{aligned}$$

The value of $\mathbb{E}[f_c]$ is deduced straightforwardly

Conclusion

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- No time to work on the Quantum Amplitude Estimation algorithm but you have enough background knowledge to work on
 - ▶ Quantum Fourier Transform
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- First tools for a *basic* quantum circuit to evaluate a call price
- No time to work on the Quantum Amplitude Estimation algorithm but you have enough background knowledge to work on
 - ▶ Quantum Fourier Transform
 - ▶ Quantum Phase Estimation
 - ▶ *Quantum Amplitude Estimation* (<https://arxiv.org/abs/quant-ph/0005055>)
- Some considerations about circuit design are worth investigating
 - ▶ How can the number of qubits and gates be reduced?
 - ▶ **Uncomputation** for an efficient reuse of quantum resources

End of the class

- The final three classes will be a hands-on lab
- Goal: design a quantum circuit in Qiskit to price a call option
- Work to be done:
 - 1 Define methods to construct a quantum circuit that encode affine functions and a circuit that compares integers
 - 2 Define a method that constructs a quantum circuit encoding a probability distribution
 - 3 Use the three defined methods and the built-in QAE component to compute the price of the call option
- Expected outcome: a Jupyter notebook **that reads as a report** and contains all the code to test the resulting circuits **and contains an analysis** of the advantages and disadvantages of using a quantum algorithm to price derivative products